BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT(NEP) **Sem: VI** **Roll No.: TYIT30** **Date : 10-12-2025**

Course Name: Cybersphere and Security(CSS) **Page no:**

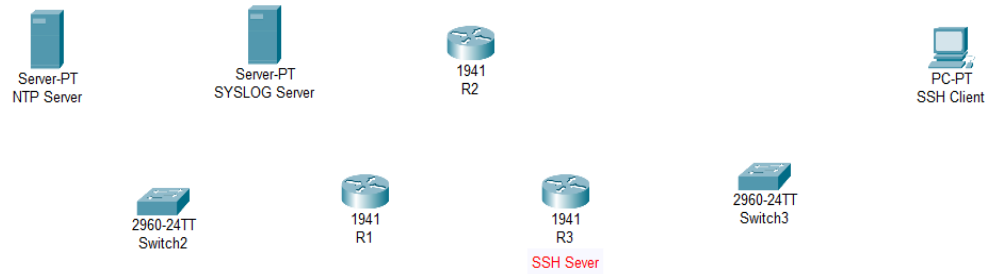
Practical Number: 2B

Q. Configure Routers: NTP

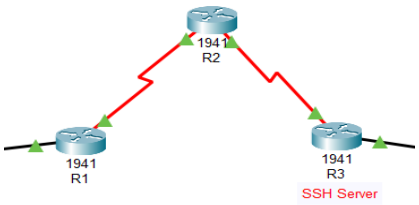
1st Steps: Setup

1. Open the Cisco packet
2. Add the three routers of 1941 (R1, R2, R3) place it like triangle.
3. Click on each router (R1, R2, R3) → Go to the "Physical" tab → Turn off the router → Drag and drop the "HWIC-2T" module into an empty slot → Turn the router back on → Repeat for all routers.

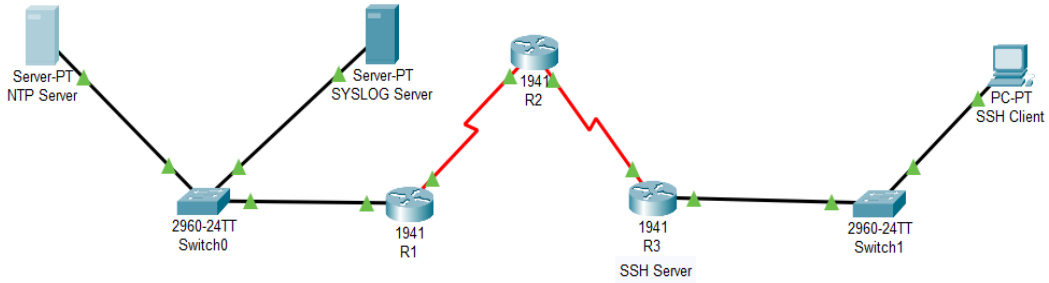
4. Take two 2960-24TT Switches → two servers → one PC and give it names as shown.



5. Connect the serial DTE wire to the router.

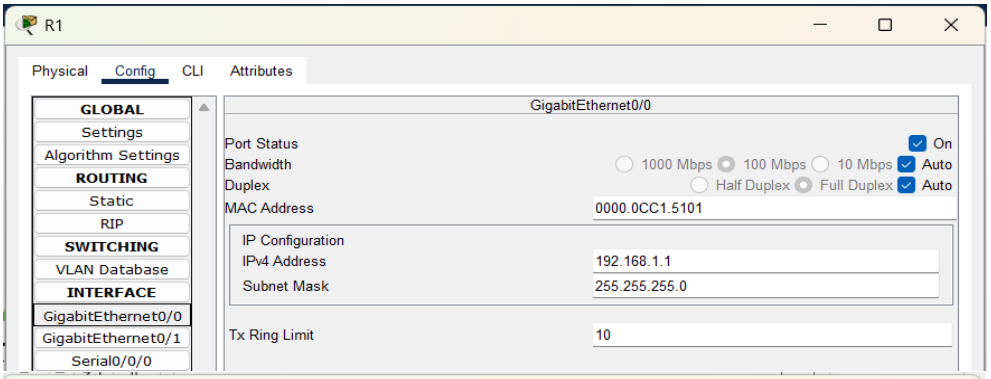


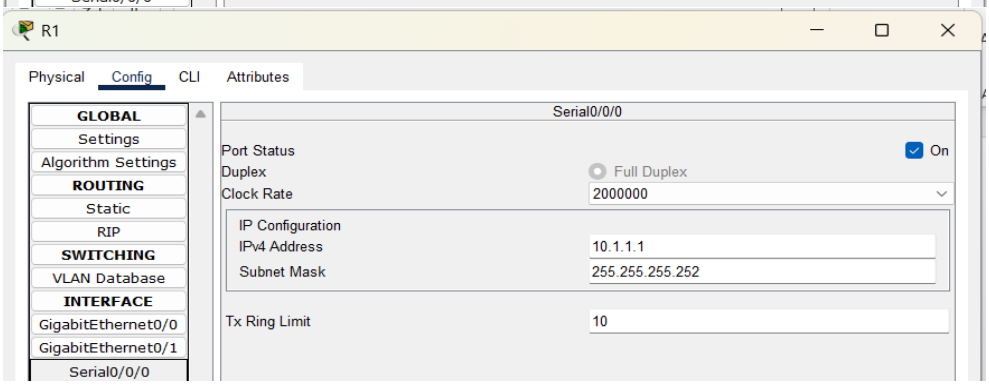
6. Connect the copper straight-through wire to devices as shown below.



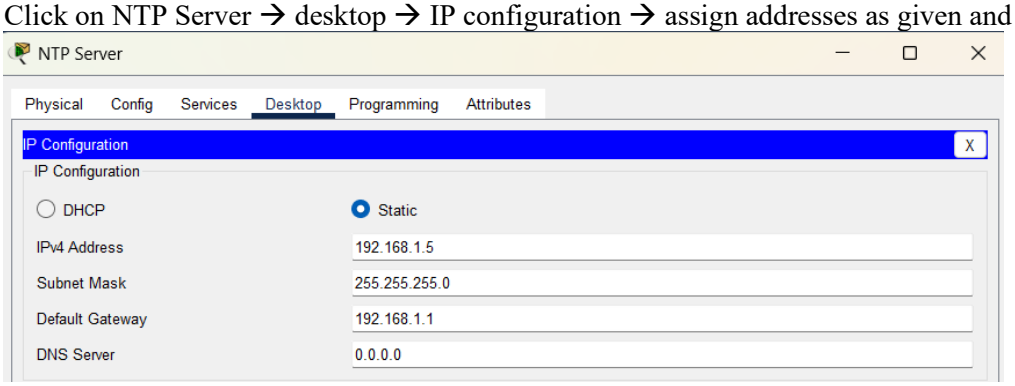
7. Click on R1 and give addresses and on port status as shown, then close the tap.

Teacher's Signature

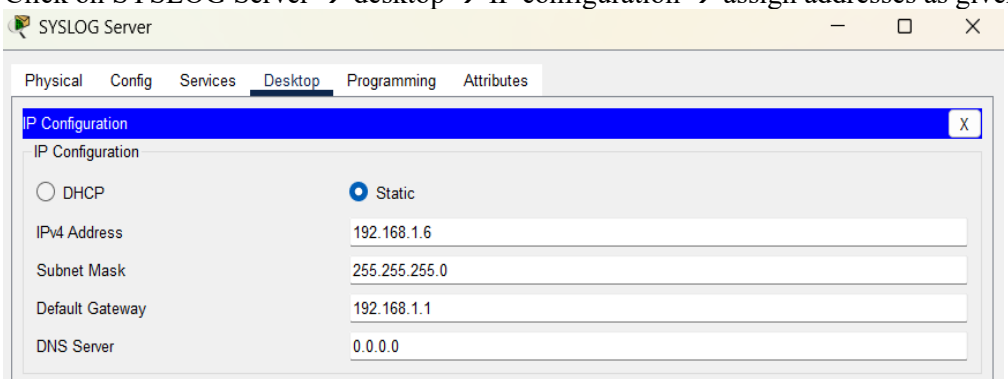




8. Click on NTP Server → desktop → IP configuration → assign addresses as given and close.



9. Click on SYSLOG Server → desktop → IP configuration → assign addresses as given and close.



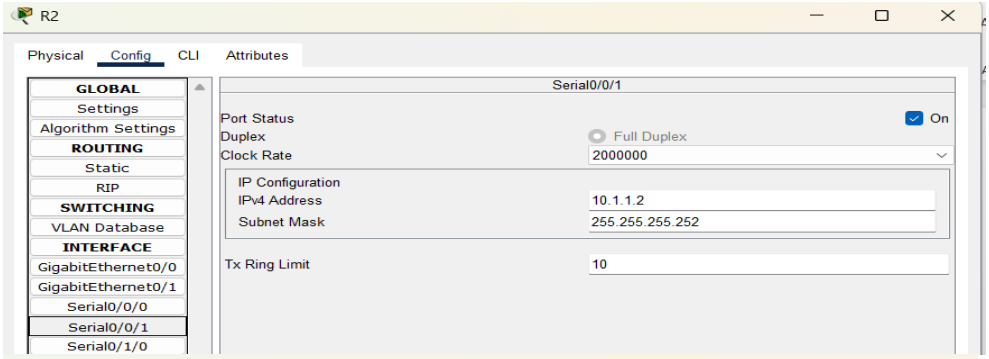
10. Click on R2 and give addresses and on port status as shown, then close the tap.

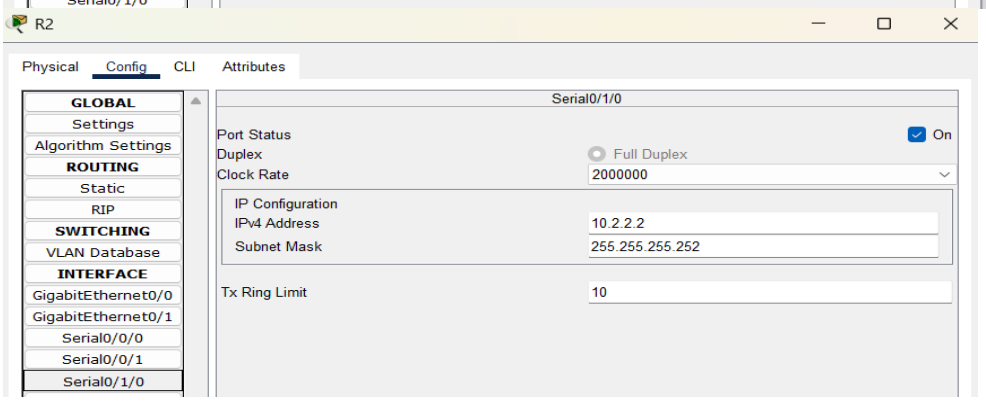
BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT(NEP) **Sem: VI** **Roll No.: TYIT30** **Date : 10-12-2025**

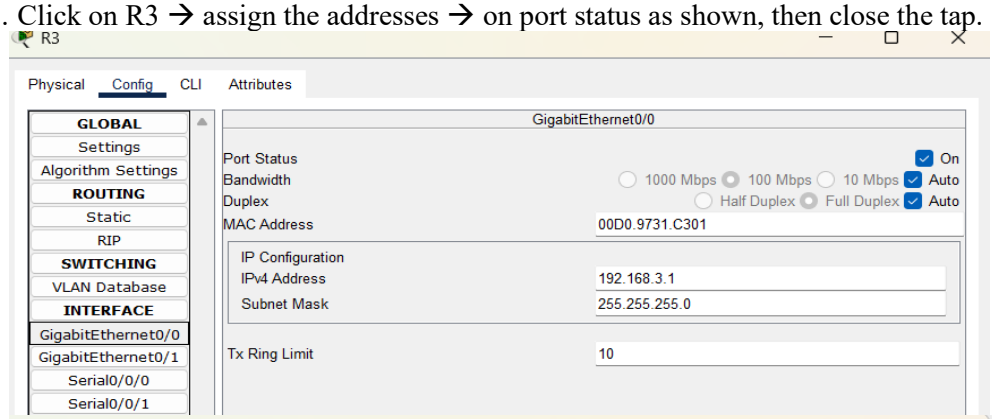
Course Name: Cybersphere and Security(CSS) **Page no:**

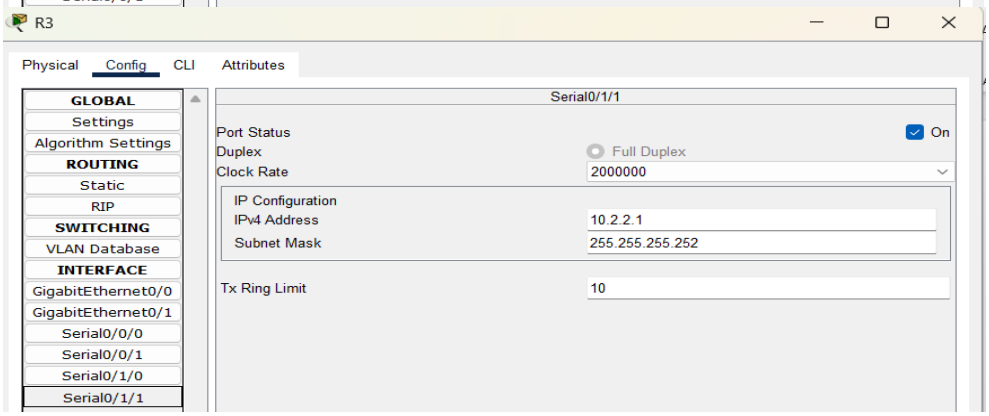
Practical Number: 2B





11. Click on R3 → assign the addresses → on port status as shown, then close the tap.





12. Click on SSH Client → desktop → IP configuration → assign addresses as given and close.

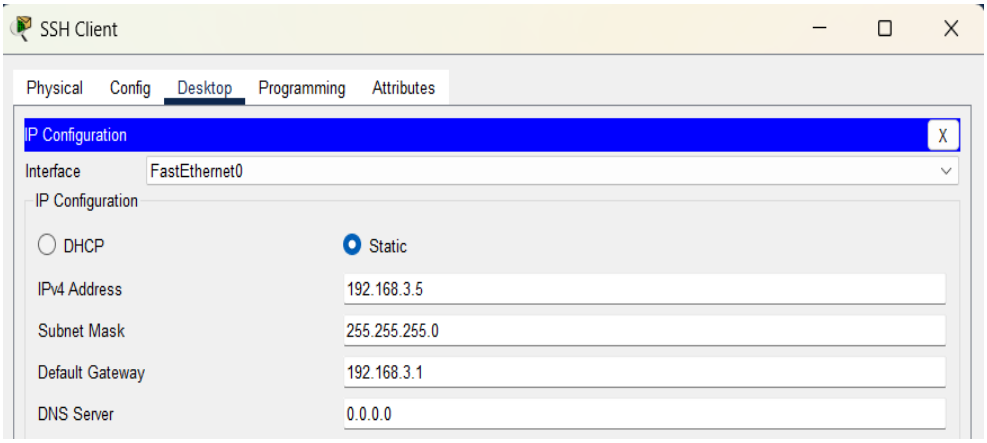
Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

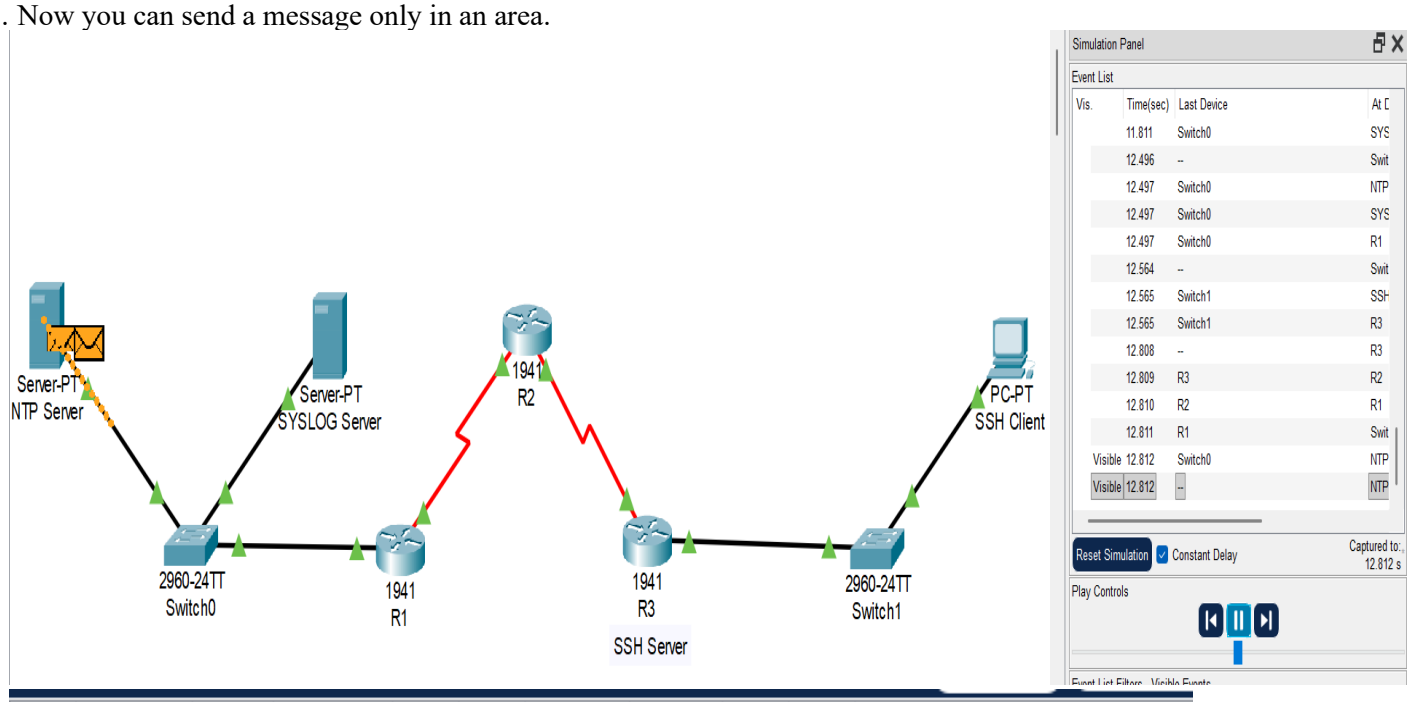
Class: TYIT(NEP) **Sem: VI** **Roll No.: TYIT30** **Date : 10-12-2025**

Course Name: Cybersphere and Security(CSS) **Page no:**

Practical Number: 2B



13. Now you can send a message only in an area.



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	NTP ...	R1	ICMP		0.000	N	0	(edit)	(delete)

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	SYSL...	R1	ICMP		0.000	N	2	(edit)	(delete)
	Successful	R3	SSH Client	ICMP		0.000	N	3	(edit)	(delete)
	Successful	SSH ...	R3	ICMP		0.000	N	4	(edit)	(delete)

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT(NEP) **Sem: VI** **Roll No.: TYIT30** **Date : 10-12-2025**

Course Name: Cybersphere and Security(CSS) **Page no:**

Practical Number: 2B

2nd Step: NTP

1. Click on R1 → CLI → paste this lines.
   - router ospf 1
   - area 0 authentication message-digest
   - interface S0/0/0
   - ip ospf message-digest-key 1 md5 MD5pa55

2. Click on R2 → CLI → paste this lines.
   - router ospf 2
   - area 0 authentication message-digest
   - interface S0/0/1
   - ip ospf message-digest-key 1 md5 MD5pa55
   - interface S0/1/
   - ip ospf message-digest-key 1 md5 MD5pa55

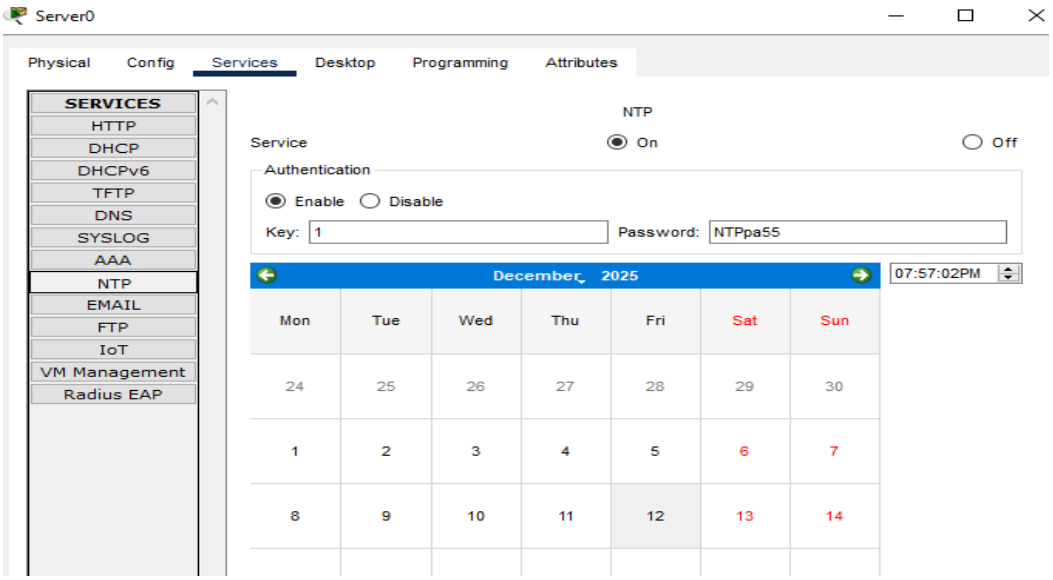
3. Click on R3 → CLI → paste this lines.
   - router ospf 3
   - area 0 authentication message-digest
   - interface S0/1/1
   - ip ospf message-digest-key 1 md5 MD5pa55

4. In R3 → CLI → paste → show ip ospf interface

```
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
      No key configured, using default key id 0
Router#
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ntp server 198.168.1.5
```

5. Click on NTP Server → Servies → NTP → if service if off then on it → authentication: Enable → key: 1 → password: NTPpa55 → then change time and date → close.

Teacher's Signature



6. Put these lines in all three routers (R1, R2, R3)
   - ntp server 198.168.1.5
   - ntp update-calendar
   - ntp authenticate
   - ntp trusted-key 1
   - ntp authentication-key 1 md5 NTPpa55
   - service timestamps log datetime msec

7. As shown below, do that to see the change.

```
Router(config)#ex
Router#
*Dec 13, 02:23:31.2323: SYS-5-CONFIG_I: Configured from console by console
Router#show clock
2:23:40.974 UTC Sat Dec 13 2025
Router#
```